Arya Vaidya

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Education

Georgia Institute of Technology — Atlanta, GA
B.S. in Computer Science. Threads: AI (Intelligence) and Devices — Expected May 2029
- Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Linear Algebra

Bridgewater-Raritan High School — Bridgewater, NJ
GPA: 3.97, Top 5% of Class
- Relevant Coursework: Calculus III, Differential Equations, AP Probability & Statistics (5)

Skills

Languages & Tools: Python, Java, C++, SQL, Git, Google Firebase, Claude Code, Cursor
Machine Learning: PyTorch, TensorFlow, OpenCV, Hugging Face, Jupyter Notebook, Transformers
App/Website Development: Swift, Xcode, React, Next.js, Stripe, HTML, Tailwind CSS

Experience

Applied AI Engineer Intern — June 2025 – Sept 2025
MoBoo — *Remote*
- Fine-tuned Llama 3.2 LLM with LoRA-based parameter-efficient training through Hugging Face to generate consolidated summaries across a tokenized corpus of 1M+ reviews
- Designed an automated evaluation pipeline using a larger Llama 3.2 LLM as a judge to score generated summaries for accuracy and faithfulness to source reviews across 35k+ books
- Built Python tools to perform BERT-based sentiment analysis on critic reviews and keyword extraction using KeyBERT, enabling 32k users to make more informed reading decisions

Instructor and Course Developer — Jan 2024 – Jul 2026
iCode — *Bridgewater, NJ*
- Taught 35+ students through project-based instruction in ML, Python, Arduino, and VEX Robotics, guiding them through hands-on capstone projects and competition-ready builds
- Built and sold a comprehensive 50-week AI curriculum to iCode covering ML, deep learning, and agentic AI, supported by original slide decks, quizzes, and hands-on Colab notebooks

Projects

ML Hypernetwork Research for Fluid Simulations — June 2025 – Present
- Engineered a lightweight hypernetwork architecture in PyTorch for multi-step fluid dynamic simulations across 4 datasets, applied transfer learning to extend models trained on 1D simulations to 2D, requiring 20× less training data
- Developed a calibrated periodic boundary-condition loss with warmup scheduling, cutting 50-step rollout MSE by 72.3%
- Co-authoring an ICLR paper on this work with Prof. Nikhil Muralidhar at the Stevens Institute of Technology

AlumHub — Apr 2025 – Present
- Built a full-stack mentorship platform using Next.js and React to connect 3k+ high school students and 2k+ alumni, backed by Firebase for authentication and SQL for data storage
- Scaled the platform to 170+ schools and 24 states by database indexes and caching to make SQL queries 3x faster, and integrated Stripe for subscription payments
- Engineered a rule-based scoring algorithm to connect students and mentors on shared interests and experiences, with an LLM re-ranking layer for the top matches

Programming Supervisor and Subcaptain for FRC Robotics — Sept 2022 – June 2026
- Managed a $50k+ budget and led 50 members to develop competition-ready robots and host 6 STEAM events
- Built a real-time object detection pipeline using YOLOv8, feeding live game piece positions into an A* pathing algorithm to autonomously navigate and score under defensive pressure
- Built a scouting app for 100+ members to log in-match robot performance data, syncing entries offline over peer-to-peer Bluetooth to a master device, enabling the scouting lead to make data-driven alliance selection picks
- Named a FIRST Leadership Award Semi-Finalist in March 2025

Generative AI Research for Art Restoration — Dec 2024 – July 2025
- Designed a latent diffusion model for artwork restoration in PyTorch combining a residual VAE with an image-conditioned diffusion transformer, achieving 0.010 L1 error on artwork damaged by random patch masking
- Validated the architecture through ablation studies against 3 baseline architectures and 6 loss function configurations, outperforming 4 state-of-the-art restoration models by 8.1% in PSNR and 1.82% in SSIM
- Authored a research paper and presented the work at the NJ ISEF Fair, placing 3rd in the Computer Science category